

# Problem Set 1

## Applied Stats/Quant Methods 1

Due: October 8, 2026

### Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

### Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113,  
        112, 98, 80, 97, 95, 111, 114, 89, 95, 126, 98)
```

*Number of Observations::*

```
1 n = length(y)
```

*Visualising the Data:*

```
1 pdf("plot_1.pdf")  
2  
3  
4 hist(y,  
5     breaks = 8,  
6     probability = TRUE,  
7     main = "IQ Scores of Students",  
8     xlab = "Scores")
```

```

9
10 lines(density(y), col = "blue")
11
12 dev.off()

```

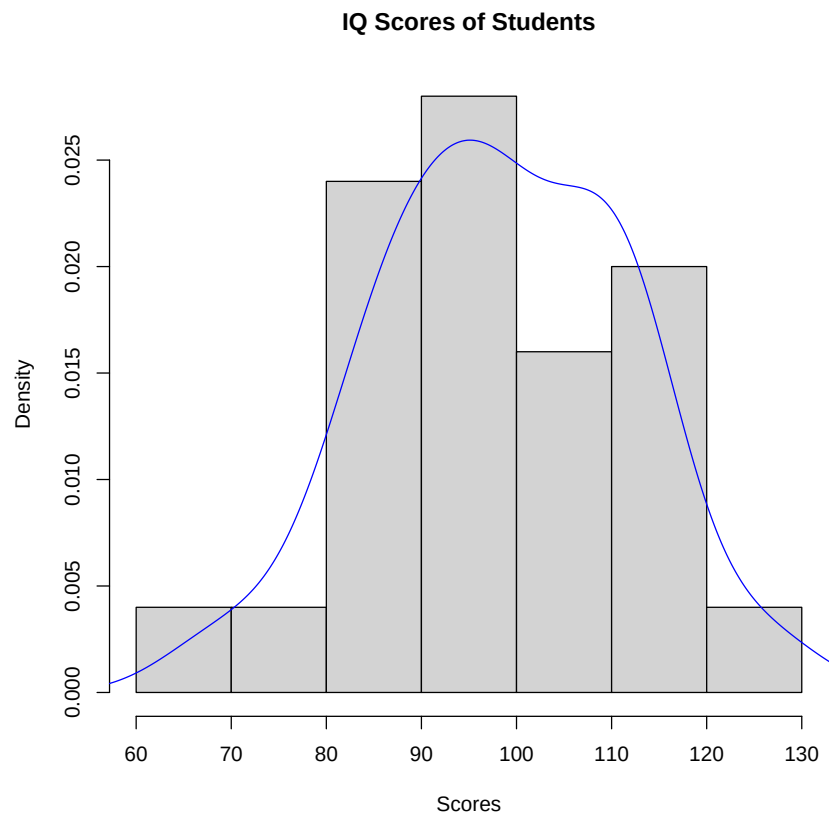


Figure 1: IQ Scores of Students

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

*Descriptive Statistics:*

```

1 mean_y <- sum(y)/n
2 sd_y <- sqrt(sum((y - mean_y) ^ 2) / (n - 1))
3 se_y <- sd_y/sqrt(n)
4 t_score <- qt(0.95, df = n - 1)
5
6 ci_90_lower = mean_y - t_score * se_y
7 ci_90_upper = mean_y + t_score * se_y

```

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Using the same sample, conduct the appropriate hypothesis test with  $\alpha = 0.05$ .

```
1 t.test(y, mu = 100, alternative = "greater", conf.level = 0.95)
```

## Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

|        |                                                                          |
|--------|--------------------------------------------------------------------------|
| State  | 50 states in US                                                          |
| Y      | per capita expenditure on shelters/housing assistance in state           |
| X1     | per capita personal income in state                                      |
| X2     | Number of residents per 100,000 that are "financially insecure" in state |
| X3     | Number of people per thousand residing in urban areas in state           |
| Region | 1=Northeast, 2= North Central, 3= South, 4=West                          |

Explore the `expenditure` data set and import data into R.

- Please plot the relationships among  $Y$ ,  $X1$ ,  $X2$ , and  $X3$ ? What are the correlations among them (you just need to describe the graph and the relationships among them)?

```
1 vars <- c("Y", "X1", "X2", "X3")
2
3 labels <- c(
4   "Expenditure",
5   "Income",
6   "Financially Insecure",
7   "Urban"
8 )
9
10 pdf("plot_2.pdf")
11
12 par(mfrow = c(2, 3))
13
14 for (i in 1:(length(vars) - 1)) {
15   for (j in (i + 1):length(vars)) {
16
17     plot(
18       expenditure[[vars[j]]],
19       expenditure[[vars[i]]],
20       xlab = labels[j],
21       ylab = labels[i],
22       main = paste(labels[i], "vs", labels[j]),
23       cex = 0.4
24     )
25
26     abline(
27       lm(expenditure[[vars[i]]] ~ expenditure[[vars[j]]]),
```

```

28     col = "blue",
29   )
30
31   r <- cor(
32     expenditure[[vars[i]]],
33     expenditure[[vars[j]]]
34   )
35
36   mtext(
37     paste("r =", round(r, 2)),
38
39   )
40 }
41 }
42
43 dev.off()

```

- Please plot the relationship between  $Y$  and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```

1 plot_3 <- ggplot(expenditure, aes(x = factor(Region), y = Y, color =
2   factor(Region))) +
3   geom_boxplot() +
4   stat_summary(
5     aes(shape = "Mean"),
6     fun = mean,
7     geom = "point",
8     size = 3,
9     alpha = 0.45
10  ) +
11  labs(
12    title = "Housing Expenditure by Region",
13    x = "Region",
14    y = "Per Capita Expenditure",
15    color = "Region",
16    shape = ""
17  ) +
18  scale_shape_manual(
19    values = c("Mean" = 18)
20  )
21 ggsave("plot_3.pdf", plot = plot_3, width = 7, height = 5)

```

- Please plot the relationship between  $Y$  and  $X1$ ? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

```

1 plot_4 <- ggplot(expenditure, aes(x = X1, y = Y)) +
2   geom_point() +
3   labs(
4     title = "Housing Expenditure vs Income",

```

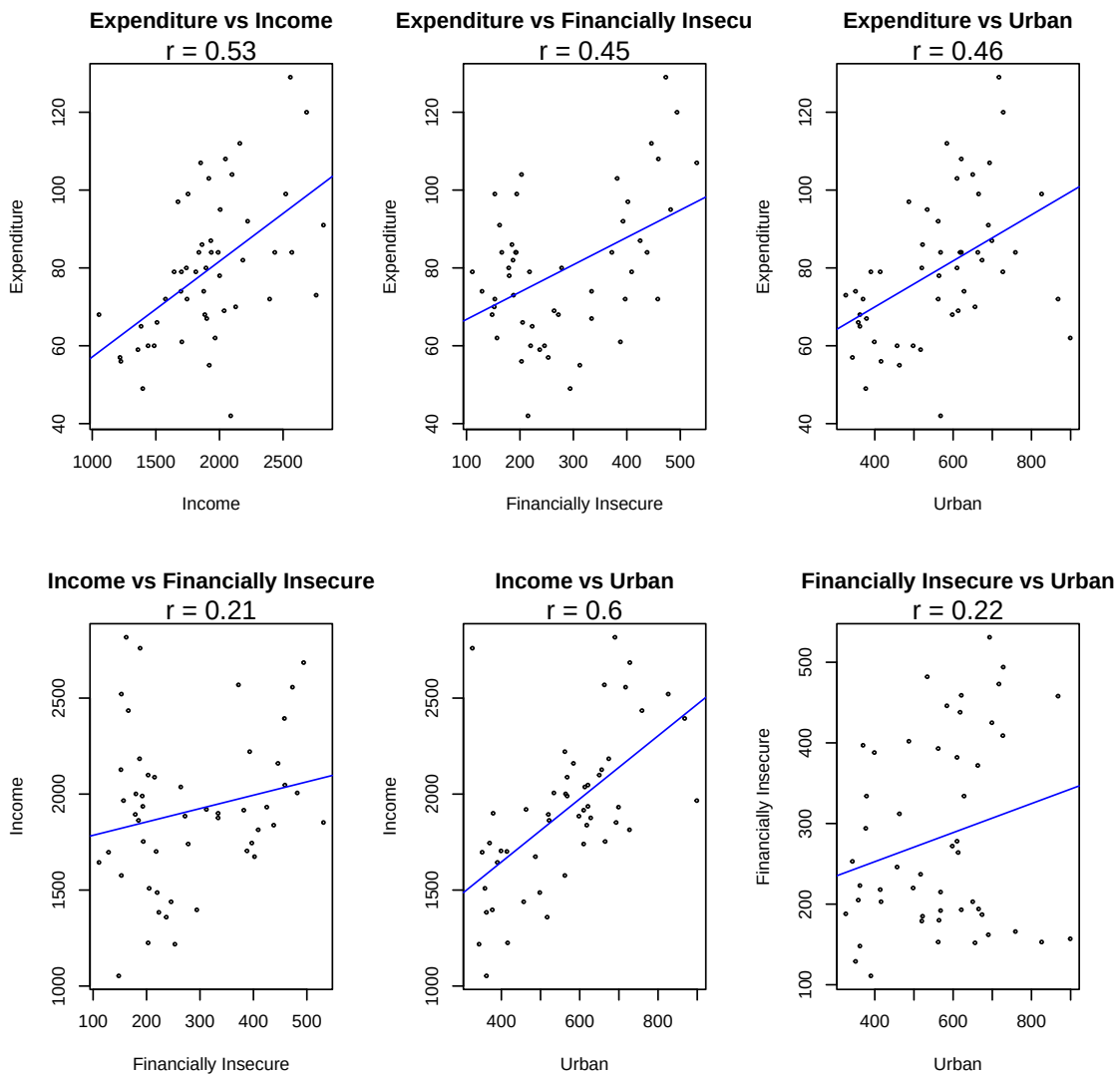


Figure 2: Pairwise Relationships Between Expenditure Variables

```

5   x = "Per Capita Income",
6   y = "Per Capita Expenditure"
7 )
8
9 ggsave("plot_4.pdf", plot = plot_4, width = 7, height = 5)

1 plot_5 <- ggplot(
2   expenditure,
3   aes(
4     x = X1,
5     y = Y,
6     color = factor(Region),

```

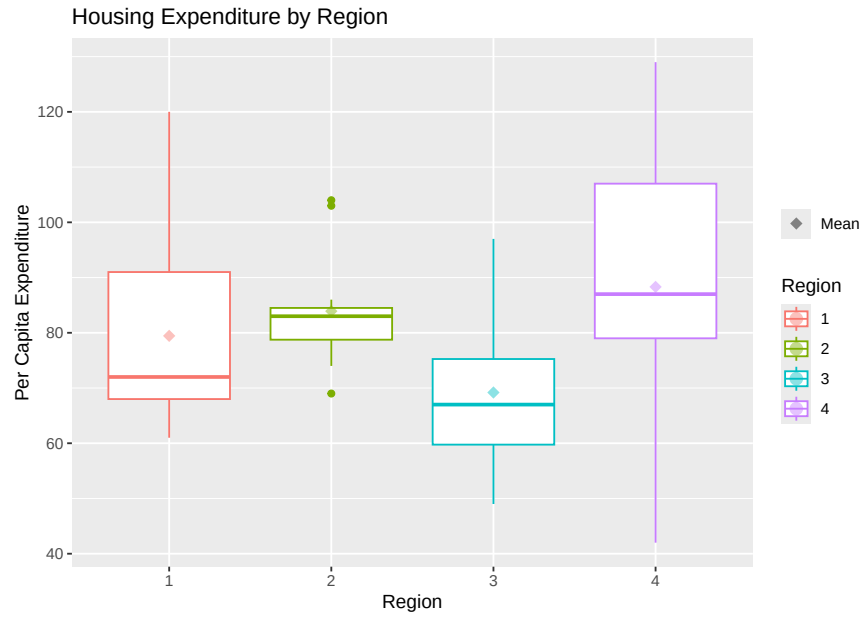


Figure 3: Housing Expenditure by Region

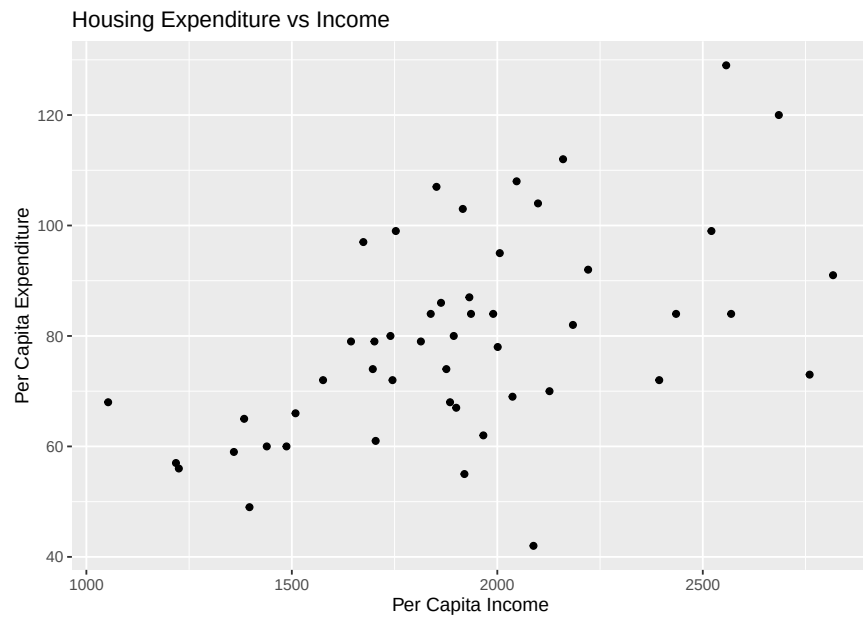


Figure 4: Housing Expenditure versus Income

```

7   shape = factor(Region)
8   )
9   ) +
10  geom_point() +
11  labs(

```

```

12     title = "Housing Expenditure vs Income by Region",
13     x = "Per Capita Income",
14     y = "Per Capita Expenditure",
15     color = "Region",
16     shape = "Region"
17 )
18
19 ggsave("plot_5.pdf", plot = plot_5, width = 7, height = 5)

```

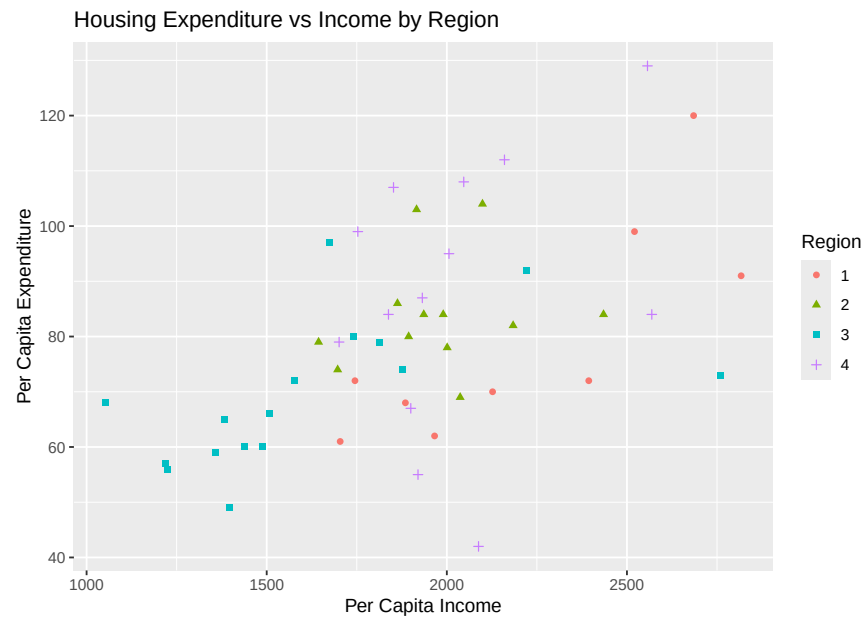


Figure 5: Housing Expenditure versus Income by Region